REDOX 1

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Redox is an abbreviation for the terms **red**uction and **ox**idation. Like many expressions in chemistry, their meanings of reduction and oxidation have evolved. Because oxygen is involved in so many common reactions (such as burning, corrosion, respiration and photosynthesis) many chemists have studied it. Oxidation originally meant the addition of oxygen to an element; reduction originally meant the removal of oxygen. Gradually, electronic theory and the concept of valency expanded, and oxidation and reduction have been related to the transfer or rearrangement of electrons. This in turn has led to the ideas of reduction and oxidation being applied to most reactions.

1 *The property of elements that determines the ratio of combining atoms of different elements.* What does this describe?

Here are some broader definitions of **oxidation**:

- The complete or partial removal of electrons from an atom
- The loss of electrons.

Here are some broader definitions of **reduction**:

- The complete or partial addition of electrons to an atom
- The gain of electrons.

There seems to be a contradiction here. Electrons are particles with a single charge – how can they be **partially** removed? The answer lies in the following descriptions:

- **Ionic bonding** involves one atom transferring one or more electrons to a different atom
- **Covalent bonding** has atoms sharing electrons
- **Polar covalent bonding** has unequal sharing of electrons.

Polar covalent bonding shows characteristics of both ionic and covalent bonding. Electrons are shared, but one atom has a greater share of the electron pair.

2 Define each type of bonding

a Ionic b Covalent c Polar covalent

3 Discuss this statement: *The processes of reduction and oxidation must take place simultaneously.*

Electronegativity This is a concept that helps in understanding redox reactions. Electronegativity is a measure of how strongly an atom attracts electrons. When two or more atoms are 'competing' for electrons, the more electronegative atoms will 'win', gain electrons and be reduced. The most electronegative elements are Group VII (the halogens) and the least electronegative elements are the Group I elements (the alkali metals).

4 What is the name of the process where an atom loses electrons?

5 Write the term that matches each description.

a An atom or molecule that readily *captures* electrons.

b An atom or molecule that readily *releases* electrons.

Reactive metals are oxidised more easily than unreactive metals. This is the reaction between sodium and chlorine: $Na + Cl_2 \rightarrow Na^+ + Cl^-$. The sodium atom has lost an electron and formed a positive ion. It has been **oxidised**. Chlorine has gained an electron and formed a negative ion; it has been **reduced**.

6 For each reaction, identify which atoms have been oxidised and which have been reduced.

a $Mg + \frac{1}{2}O_2 \rightarrow Mg^{2+} + O^{2-}$

b $Zn + S \rightarrow Zn^{2+} + S^{2-}$

c $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$

7 Which Group of elements contain the most electronegative elements?

8 Which Group of elements contain the least electronegative elements?

9 Name three types of reactions that involve oxygen.

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Oxidation number (state)

The oxidation number of an atom is related to its **valency**. (Valency tells the number of electrons involved in the bonding of a particular atom.) When we work out oxidation numbers, we assume that all bonds involve the complete transfer of electrons. The oxidation number tells whether an atom has gained or lost control of electrons and also the number of electrons involved. Because electrons have a negative charge, an atom that loses electrons will have a positive oxidation number; an atom that gains electrons will have a negative oxidation number. Note that the terms **oxidation number** and **oxidation state** are usually regarded as referring to the same thing.

The following rules apply in regard to oxidation numbers:

- All bonding between different types of atoms is regarded as **ionic bonding**
- Atoms in the elementary state (i.e. not bonded with other elements) are assigned an oxidation number of zero
- A positive sign (+) or a negative sign (–) is put before the oxidation number
- A positive oxidation number means an atom has lost electrons
- A negative oxidation number means an atom has gained electrons
- The oxidation number of an ion is the same as its ionic charge number
- Hydrogen has an oxidation state of +1 when bonded with non-metals
- Oxygen usually has an oxidation number of –2 except in peroxides
- Group I metals have an oxidation number of +1
- Group II metals have an oxidation number of +2
- Group VII elements usually have an oxidation state of –1 when bonded with metals
- Transition metals have variable oxidation states
- The algebraic sum of the oxidation numbers of atoms in a compound is zero
- If an atom has more than one possible oxidation number, the actual number is shown by Roman numerals in brackets immediately after the name of the atom in a formula. For example, in Cu (II), copper is in the +2 oxidation state.

We can use these rules to calculate the oxidation number of a particular element in a compound.

EXAMPLES

i **What is the oxidation state of oxygen in hydrogen peroxide?**

Hydrogen peroxide has the formula H_2O_2.

The overall charge is zero, so the oxidation numbers for each element (hydrogen and oxygen) must add up to zero.

The oxidation number of hydrogen is +1.

There are two hydrogen atoms, so the sum of the oxidation number for hydrogen is $(2 \times +1) = +2$.

This means that oxidation numbers for oxygen must equal –2. There are two oxygen atoms, so each individual oxygen atom must have an oxidation number of $(-2 \div 2) = -1$.

ii **What is the oxidation state of manganese in potassium permanganate?**

Potassium permanganate has the formula $KMnO_4$.

The overall charge is zero, so the oxidation numbers of potassium, manganese and oxygen atoms must add up to zero.

The oxidation number of potassium is +1. (Potassium is a Group I metal.) There is one potassium atom, so the oxidation number for potassium atoms is: $1 \times +1 = +1$.

Manganese is a transition metal, so the oxidation number is not immediately apparent.

The oxidation number of oxygen is –2. There are four oxygen atoms, so so the oxidation number for oxygen atoms is: $4 \times -2 = -8$.

Add up the oxidation numbers: K = +1, Mn = ?, $O_4 = -8$. $+1 + ? + -8 = 0$; hence Mn = +7.

10 Calculate the oxidation number of each given element. Show your working.

a Nitrogen in ammonia (NH_3)
b Nitrogen in nitric acid (HNO_3)
c Sulfur in hydrogen sulfide (H_2S)
d Sulfur in sulfuric acid (H_2SO_4)

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